DEFORMABLE ALIGNMENT AND SCALE-ADAPTIVE FEATURE EXTRACTION NETWORK FOR CONTINUOUS-SCALE SATELLITE VIDEO SUPER-RESOLUTION

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ABSTRACT

Video super-resolution (VSR), especially continuous-scale VSR, plays a crucial role in improving the quality of satellite video images. Continuous-scale VSR aims to use a single model to process arbitrary (integer or non-integer) scale factors, which is conducive to meeting the needs of video images transmission with different compression ratios and arbitrarily zooming by rolling the mouse wheel. In this article, we propose a novel network to achieve continuous-scale satellite VSR (CAVSR). Specifically, first, we propose a time-series-aware dynamic routing deformable alignment module (TDAM) for feature alignment between frames at different times. Second, we develop a scale-adaptive feature extraction module (SFEM), which uses the proposed scale-adaptive convolution (SA-Conv) to generate dynamic scale-adaptive filters based on the input scale information. Finally, we design a global implicit function feature-adaptive walk continuous-scale upsampling module (GFCUM), which can perform different feature walks according to the input features with different scale information and finally complete the continuous-scale mapping from coordinates to pixel values. Experimental results have demonstrated the CAVSR has superior reconstruction performance.

***Index Terms*—** Video super-resolution, continuous-scale, deformable convolutional alignment, scale-adaptive feature, feature-adaptive walk

1. INTRODUCTION

High-resolution (HR) satellite video with rich texture is more preferred to humans and has been widely used in object tracking [1] and land-cover classification [2]. However, to improve the transmission speed and efficiency, satellite video usually sacrifices its spatial resolution, which hinders its application. Therefore, the super-resolution (SR) technique is necessary to apply to satellite video, which aims to recover HR images from the corresponding low-resolution (LR) images [3, 4]. Different from single image SR, which only uses the spatial information of the single LR image. VSR is more complex, which aims to reconstruct an HR video frame from its corresponding LR video frame (reference frame) and multiple neighboring LR video frames (supporting frames).

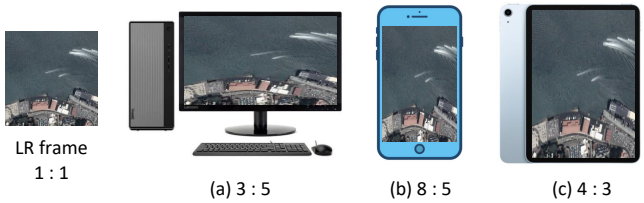

Fig. 1. The results of asymmetrical satellite VSR. (a) Apply to computer. (b) Apply to mobile phone. (c) Apply to ipad.

VSR needs to consider both spatial and temporal information. Therefore, the key to VSR is how to process the redundant information among frames.

Deep learning technology has demonstrated its prominent superiority in the field of VSR [5, 6, 7]. Many methods [8, 9, 10] based on deep learning have been proposed to achieve VSR. They use optical flow to predict the motion field between the reference frame and the supporting frame, and then wrap the supporting frame with their corresponding motion fields. However, incorrect optical flow calculations may introduce artifacts to reconstructed frames. Therefore, Tian *et al* [11] and Wang *et al* [12] proposed a module based on deformable convolution to achieve temporal feature alignment, which improved performance. But, they only consider using a unified module to align all supporting frames, without considering the differences among them. In addition, the resolution of satellite video frames is lower than that of natural video and lacks sufficient texture and detail information, making feature extraction more difficult. Furthermore, moving objects have different scales and occupy very few pixels in satellite video frames, which makes them difficult to capture. More important, previous methods can't achieve continuous-scale VSR, which limits the application of satellite video.

To solve the above problems, this article proposed a network for continuous-scale satellite VSR with deformable alignment and scale-adaptive feature extraction. Furthermore, our network can achieve asymmetrical VSR, which makes it a reality for videos to be displayed on different monitors without black borders. The results of asymmetrical VSR are shown in Fig. 1. The main contributions are as follows:

1) We propose a time-series-aware dynamic routing deformable alignment module (TDAM). The TDAM can choose different reconstruction paths for different frames at different times, which can effectively learn more complex

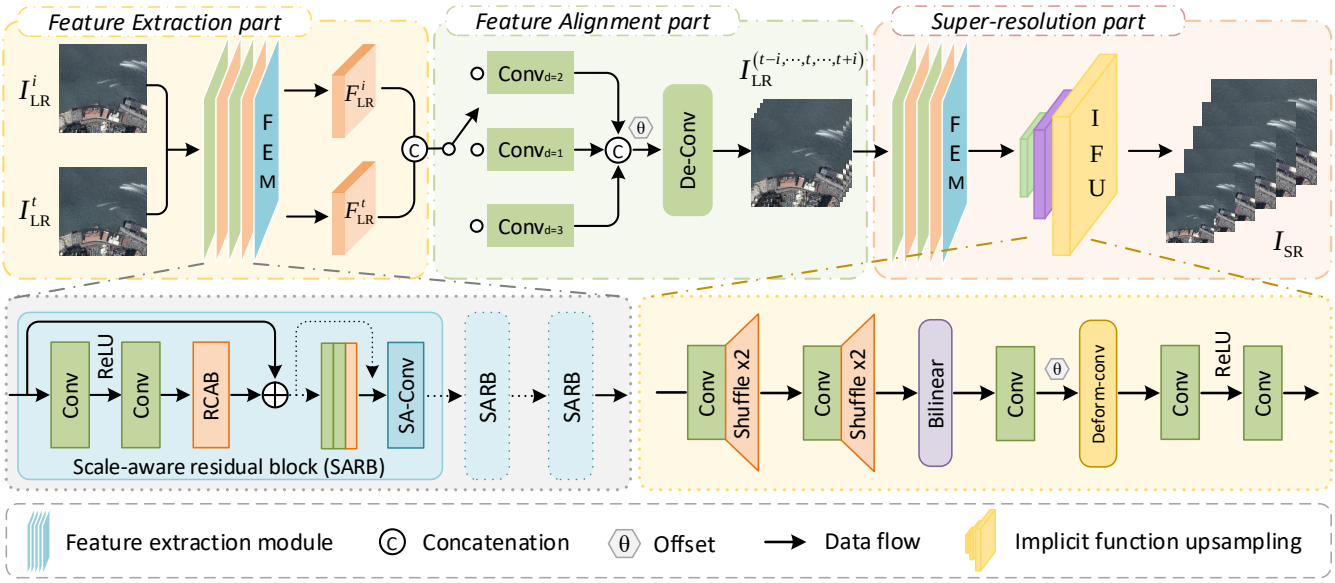

Fig. 2. The flowchart of the proposed CAVSR.

motion information in satellite video and complete the effective fusion of spatial-temporal information.

2) We proposed a scale-adaptive feature extraction module (SFEM). The SFEM contains the SA-Conv. The SA-Conv can dynamically adjust the filter weights according to the input scale information, which allows the SFEM to better adapt to the multi-task learning of various scale factors.

3) A global implicit function feature-adaptive walk continuous-scale upsampling module (GFCUM) is designed for continuous-scale VSR. The GFCUM can perform different feature walks according to the input features with different scale information and finally complete the continuous-scale mapping from coordinates to pixel values.

2. METHODOLOGY

In this section, we elaborate on the architecture and loss function of the CAVSR.

2.1. Overview

As shown in Fig. 2, our network CAVSR consists of three main parts: the feature extraction part (FEP), the feature alignment part (FAP), and the super-resolution part (SRP). The FEM in Fig. 2 is the proposed SFEM and the IFU in Fig. 2 is the proposed GFCUM. Let $I^t_{\mathrm{LR}} \in R^{H\times W\times c}$ be the tth LR video reference frame and $I^t_{\mathrm{SR}} \in R^{r*H\times r*W\times c}$ be the VSR frame, where r is the upscaling factor, c = 3 represents the RGB channels, and H and W represent the height and width of the LR frame, respectively. Given a group LR video frames, including reference frame I^t_{LR} and $2N$ supporting frames $\{I^{t-N}_{\mathrm{LR}},\cdots,I^{t-1}_{\mathrm{LR}},I^{t+1}_{\mathrm{LR}},\cdots,I^{t+N}_{\mathrm{LR}}\}$. We first extract the features F^t_{LR} of reference frame I^t_{LR} as

$$F^t_{\mathrm{LR}} = f_{\mathrm{FEP}}(I^t_{\mathrm{LR}}), \tag{1}$$

where f_{FEP} denotes the operations of FEP. We can get a group features $\{F^{t-N}_{\mathrm{LR}},\cdots,F^{t-1}_{\mathrm{LR}},F^{t+1}_{\mathrm{LR}},\cdots,F^{t+N}_{\mathrm{LR}}\}$ by doing the same operations for all supporting frames.

Second, the FAP takes the reference frame feature F^t_{LR} and a supporting frame feature F^i_{LR} as inputs to predict the corresponding aligned frame $I^i_{\mathrm{LR'}}$

$$I^i_{\mathrm{LR'}} = f_{\mathrm{FAP}}(F^t_{\mathrm{LR}}, F^i_{\mathrm{LR}}), \tag{2}$$

where f_{FAP} denotes the operations of FAP. The value range of i is $[t-N, t-1]$ and $[t+1, t+N]$, i is an integer and we can obtain 2N corresponding aligned LR frames $\{I^{t-N}_{\mathrm{LR'}},\cdots,I^{t-1}_{\mathrm{LR'}},I^{t+1}_{\mathrm{LR'}},\cdots,I^{t+N}_{\mathrm{LR'}}\}$.

Finally, The SRP will use the 2N aligned frames along with the reference frame to reconstruct the SR video frame

$$I^t_{\mathrm{SR}} = f_{\mathrm{SRP}}(I^{t-N}_{\mathrm{LR'}},\cdots,I^{t-1}_{\mathrm{LR'}},I^t_{\mathrm{LR}},I^{t+1}_{\mathrm{LR'}},\cdots,I^{t+N}_{\mathrm{LR'}}), \tag{3}$$

where f_{SRP} denotes the operations of SRP.

The network is optimized by the L_1-based loss function. Two loss functions are considered to train the network CAVSR. One is the alignment loss between supporting frames and the reference frame. Another is reconstruction loss between the reconstructed frame and the corresponding HR video frame. The specific definition is as follows:

$$L = \left\|I^t_{\mathrm{SR}} - I^t_{\mathrm{HR}}\right\|_1 + \frac{1}{2N}\sum_{i=t-N,\neq t}^{t+N}\left\|I^i_{\mathrm{LR'}} - I^t_{\mathrm{LR}}\right\|_1, \tag{4}$$

where I^t_{HR} is the corresponding HR video frame.

2.2. TDAM

We think the longer time distance from the reference frame, the more difficult to complete the alignment of supporting frames. Furthermore, the sparseness and different scales of

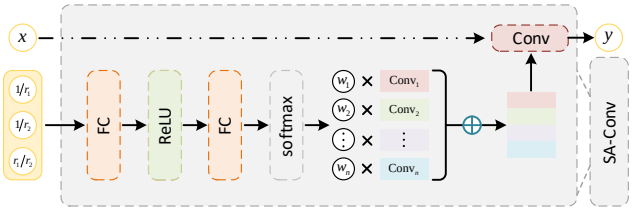

Fig. 3. The structure of the SA-Conv.

moving objects in satellite video frames, which make it difficult to capture. To alleviate the above problems, inspired by [11] and [13], we propose a TDAM. As shown in FAP of Fig. 2, our TDAM can make frames at different times through different paths. There are three paths in TDAM, the main path is a convolution operation of dilation = 1, which mainly captures normal-scale moving targets, and the two auxiliary paths are convolution operations of dilation = 2 and dilation = to assist in capturing large-scale moving targets. In addition, when the distance between the supporting frame and the reference frame is big, the targets in satellite video frames are moving far. Therefore, the wide receptive field of the two auxiliary paths can help to complete the frame alignment. The specific processes are as follows. We first take the F^t_{LR} and F^i_{LR} to obtain the offset of the filters

$$\theta^i = \begin{cases} Conv_{d=1}(F^t_{\text{LR}}, F^i_{\text{LR}}), i = t-1, t+1 \\ Conv_{d=1,2,3}(F^t_{\text{LR}}, F^i_{\text{LR}}), i = t-2, t+2 \end{cases}, \quad (5)$$

where $Conv_{d=1}$ denotes the operation of main path and $Conv_{d=1,2,3}$ denotes the operations of three paths. Specifically, $\theta^i = \{\Delta p_k | k \in \Re\}$, when the filter is 3×3, the $\Re = \{(-1,-1),(-1,0),\cdots,(0,1),(1,1)\}$ denotes nine sampling positions of the filter. We use the deformable convolution to complete alignment with the offset θ^i and the feature F^i_{LR}

$$F^i_{\text{LR}'} = Deconv(F^i_{\text{LR}}, \theta^i). \quad (6)$$

Specifically, the $F_{\text{LR}'}$ value at position p can be calculated by

$$F^i_{\text{LR}'}(p) = \sum_{p_k \in \Re} w(p_k) \cdot F^i_{\text{LR}}(p + p_k + \Delta p_k). \quad (7)$$

where $w(p_k)$ represents the weight of filter at position p_k.

2.3. SFEM

When achieving SR tasks with different scale factors, the desired features are different. Therefore, to adapt SFEM to multi-scale factors learning and inspired by [14], we propose the SA-Conv, which structure is shown in Fig. 3. As shown in Fig. 2, the SFEM consists of M scale-aware residual block (SARB). The SARB contains multi-level residual learning, which can improve convergence speed and fitting performance and finally through an SA-Conv. Through the SFEM, we can obtain the scale-adaptive feature

$$F_{\text{out}} = f_{\text{SFEM}}(I^{t-N}_{\text{LR}'}, \cdots, I^{t-1}_{\text{LR}'}, I^t_{\text{LR}}, I^{t+1}_{\text{LR}'}, \cdots, I^{t+N}_{\text{LR}'}), \quad (8)$$

where f_{SFEM} denotes the operations of SFEM. The SARB first uses two convolutional layers to enhance the receptive field.

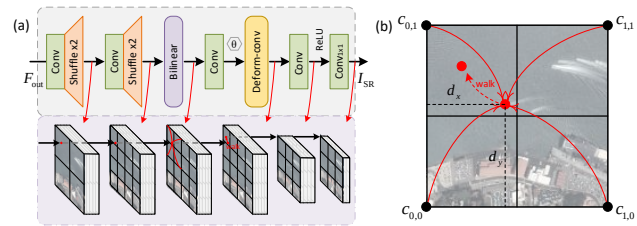

Fig. 4. (a) The structure of the GFCUM. (b) An illustration of the local implicit function and feature-adaptive walk.

Then, we use channel attention block RCAB [15] to refine features. The core idea of the SA-Conv is using a multilayer perceptron (MLP) to adaptively adjust the weights of the K parallel convolution kernels according to the input scale factors r_1 and r_2 to calculate itself parameters $\mathbf{W}$ and $\mathbf{b}$

$$\begin{aligned} \mathbf{W} &= f_{\text{MLP}}(1/r_1, 1/r_2, r_1/r_2) \cdot \mathbf{W}_i, \\ \mathbf{b} &= f_{\text{MLP}}(1/r_1, 1/r_2, r_1/r_2) \cdot \mathbf{b}_i, \end{aligned} \quad (9)$$

where $\mathbf{W}_i$ and $\mathbf{b}_i$ represent the weights and biases of the parallel convolution kernels respectively. r_1 and r_2 represent video frames SR scale factors in length and width directions respectively. f_{MLP} represents the operation of MLP.

2.4. GFCUM

The implicit function is a continuous and differentiable function parameterized by a neural network, which has been successfully applied in 3D tasks [16]. The proposed GFCUM can perform feature-adaptive walk and complete the continuous-scale mapping from coordinates to pixel values and its structure is shown in Fig. 4 (a). The GFCUM includes multiple identical local implicit functions, and we can obtain the VSR frame I^t_{SR} by

$$I^t_{\text{SR}} = f_{\text{GFCUM}}(r_1, r_2, b(F^{c_1}_{\text{out}}, C_1), \cdots, b(F^{c_n}_{\text{out}}, C_n)), \quad (10)$$

where f_{GFCUM} denotes the operations of GFCUM, r_1 and r_2 are scale factors, and b is the bilinear and represents local implicit function. $C_t = \{c_{0,0}, c_{0,1}, c_{1,0}, c_{1,1}\}$ is the coordinates and $F^{c_t}_{\text{out}}$ denotes the feature calculated at coordinates C_t.

Our GFCUM first use sub-pixel [17] layer to complete the initial VSR. Then, the local implicit function b is used to calculate the feature at coordinates C_t, and it takes the form:

$$b(F^{c_t}_{\text{out}}, C_t) = [1 - d_x, d_x] \begin{bmatrix} c_{0,0} & c_{0,1} \\ c_{1,0} & c_{1,1} \end{bmatrix} \begin{bmatrix} 1 - d_y \\ d_y \end{bmatrix}, \quad (11)$$

where (d_x, d_y) denotes the coordinate of $F^{c_t}_{\text{out}}$ relative to $c_{0,0}$. After bilinear operation, we can obtain the feature F^{r_1,r_2}_{out}.

Second, we think the bilinear operation is imprecise and unreliable. Therefore, in order to improve the accuracy and reliability of VSR. Given the feature F^{r_1,r_2}_{out} with the information of scale factors r_1 and r_2, we use deformable convolutional to finish the feature-adaptive walk

$$\begin{aligned} O_{walk} &= Conv(F^{r_1,r_2}_{\text{out}}), \\ F^{walk}_{\text{out}} &= Deconv(F^{r_1,r_2}_{\text{out}}, O_{walk}), \end{aligned} \quad (12)$$

Table 1. Quality comparison (PSNR/SSIM) for continuous-scale fastors on the dataset Set001.

Methods	×2.3	×2.6	×2.9
Bicubic	38.85/0.9692	37.09/0.9570	35.57/0.9422
CAVSR	41.27/0.9793	40.49/0.9745	39.53/0.9682
Methods	×3.0	×3.8	×4.0
Bicubic	35.15/0.9376	33.48/0.9146	31.57/0.8801
CAVSR	39.14/0.9657	37.75/0.9547	35.87/0.9360

where F_{out}^{walk} denotes the feature after walking. An illustration of the local implicit function and feature-adaptive walk is shown in 4 (b).

Finally, the GFCUM finash the mapping from feature F_{out}^{walk} to VSR frame I_{SR}^{t} after going through two convolutional layers

$$I_{\text{SR}}^{t} = Conv_{1\times1}(Relu(Conv_{3\times3}(F_{\text{out}}^{walk}))). \quad (13)$$

3. EXPERIMENTS

3.1. Datasets and Evaluation Metrics

We have nine scenes of video data from Jilin-1 satellites, which have a resolution of $1m$. For training, we crop the video into scenes from six satellite videos with a size of 640 $\times$ 640 and only consider the first 105 frames, the remaining frames are used to make test datasets. Finally, we obtain one training dataset and three test datasets (Set001-Set003). To prove the CAVSR performance objectively, we introduce two reference metrics: the peak signal-to-noise ratio (PSNR) and the structural similarity index measure (SSIM) [18].

3.2. Training Details

Set supporting frames $2N = 4$ and convolutional kernels of SA-Conv $K = 4$, the FEM of FEP contains $M = 1$ SARB and the FEM of SRP contains $M = 3$ SARB. During training, LR frames are obtained by the corresponding HR frames through the bicubic method, we randomly crop LR patches with a size of 48 $\times$ 48 as input and set the batch size to 32. In addition, We randomly flip frames vertically or horizontally and randomly rote them 90° for data augmentation. The initial learning is 1×10^{-4} and decreases by $3/4$ every 50 epochs. The r_1 in the training phase are distributed from 1 to 4. The $r_2 = r_1 + rand(0, 0.5)$ when $r_1 \in [2.5, 3.5]$ & $rand(0, 1) > 0.8$, $r_2 = r_1 - rand(0, 1)$ when $r_1 = 4.0$ & $rand(0, 1) > 0.7$, and others $r_2 = r_1$. Besides, we retrain all of the comparative networks with the same experimental settings.

3.3. Experiment Results

According to Table 1 and Table 2, we can find that our CAVSR significantly outperforms Bicubic on continuous-scale factors, and even compared to the best comparison algorithm TDAN on scale factor ×4, the PSNR is nearly improved by 0.3 dB. In addition, as shown in Fig. 5, we can conclude that our CAVSR can generate sharper and more pleasing textures. Furthermore, we also verify the effectiveness and rationality of the CAVSR structure through the ablation experiments, which can be found in Table 3.

Table 2. Quality comparison (PSNR/SSIM/Time) for scale fastor ×4 on the dataset Set002 and Set003, bold is best.

Methods	Set002	Set003
Bicubic	29.42/0.8624/0.072	29.06/0.8513/0.071
TDAN[11]	34.34/0.9301/**0.052**	33.64/0.9221/**0.051**
EDVR[12]	33.27/0.9307/0.058	32.97/0.9246/0.058
SOFVSR[19]	33.65/0.9194/0.080	33.09/0.9131/0.081
CAVSR	**34.83/0.9364**/0.055	**34.17/0.9293**/0.055

Table 3. Ablation experiments (PSNR/SSIM) on the Set002 and Set003 datasets. Rout represents the dynamic routing. Walk represents the feature adaptive-walk.

Rout	Walk	SA-Conv	Set002	Set003
×	×	×	34.17/0.9275	33.47/0.9194
×	✓	✓	34.76/0.9358	34.13/0.9290
✓	×	✓	34.81/0.9363	34.14/0.9293
✓	✓	×	34.28/0.9289	33.63/0.9212
✓	✓	✓	**34.83/0.9364**	**34.17/0.9293**

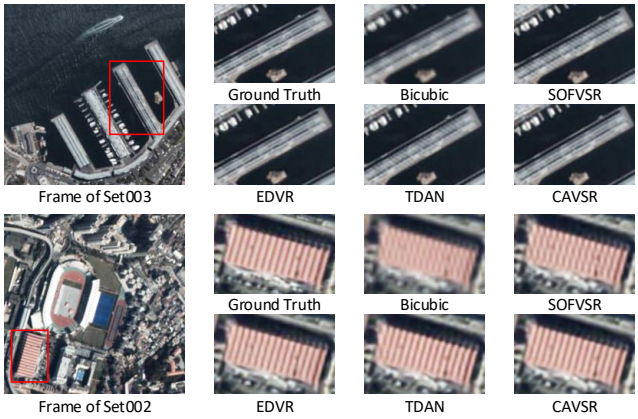

Fig. 5. The comparison of visual results for scale factor ×4.

4. CONCLUSION

In this paper, the network CAVSR is proposed to achieve continuous-scale satellite VSR with a single model. We propose a TDAM, which can achieve dynamic routing temporal feature alignment for different frames at different times. The SFEM and SA-Conv are proposed for multi-task learning of various scale factors. Finally, we propose a GFCUM, which can perform different feature walks according to the input features with different scale information and finally complete the continuous-scale mapping from coordinates to pixel values. Extensive experimental results fully demonstrate the superiority and rationality of our proposed network CAVSR.

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